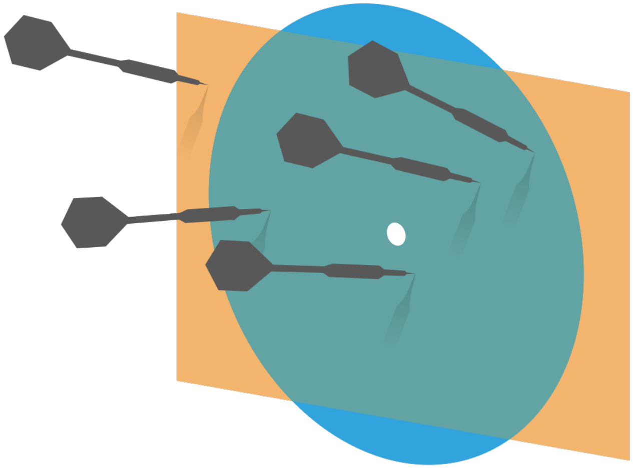
Probability and Statistics for Computer Science



Credit: wikipedia

"The eternal mystery of the world is its comprehensibility … The fact that it is comprehensible is a miracle." – Albert Einstein

Hongye Liu, Teaching Assistant Prof, Course CS361, UIUC, 08.26.2024

Objectives

- Welcome/Orientation
  - Expectations in lectures
- Big picture of the contents
- Lecture 1 - Data Visualization & Summary (I)

What to expect about the lectures?

In person + lecture
Recordings + digital notes

Poll Everywhere

Activities using **Poll Everywhere**
and or **on Canvas**

CANVAS

What to expect in the lecture?

- Students are expected to stay through the whole lecture. Early exits are only permitted for given proper reasons including accommodation
- Questions will be collected by the assistant for answers or summary.

Vision (PCA)

- **P**assion for learning
- **C**ompassion for each other
- **A**uthentic understanding

How to succeed in this course?

- Factors that will hinder you from success
- Factors that will help you succeed

Avoid these that could cause failure

- Academic integrity infraction – by all means!
- Missing homeworks, project or quizzes
- Late/Poor homeworks or project
- Insufficient viewing of the contents
- Poor time management & Procrastination
- Too many challenging classes at the same time
- Not motivated/not interested in the topic

Factors that will help you succeed

- **Be engaged/motivated**,
- **Do not hesitate to ask** for help.
- Be **Active** in class participation
- Do as much practice as possible, not just the homework and project.
- Participate in the optional teamwork
- Clear your doubts/misconceptions **asap (every lecture/discussion is important)**

Interactions are important!

- Office hours are there for you, extra points upto 3pts.
- Try to meet or talk to the instructor as least once personally
- You are encouraged to join the teamwork
- Show compassion via community service

Syllabus

- Details are on Canvas website.
  - Use the tab under the banner
- Orientation quiz is worth 8pts, you have 10 trials.

Graded Group discussion

Benefit of graded group discussion

- 30 (13%) students improved with half letter grade
- No penalty if you missed some of them

A to A+	7
A- to A	6
B+ to A-	5
B to B+	4
B- to B	4
C+ to B-	1
C to C+	1
C- to C	1
D+ to C-	1

Benefit of graded group discussion

- 57 (17.5%) students improved with half letter grade
- No penalty if you missed some of them

A to A+	7
A- to A	13
B+ to A-	12
B to B+	8
B- to B	3
C+ to B-	3
C to C+	4
C- to C	5
D+ to C-	2

Benefit of graded group discussion

- 65 (20%) students improved with half letter grade
- No penalty if you missed some of them

A to A+	15
A- to A	14
B+ to A-	15
B to B+	7
B- to B	9
C+ to B-	2
C to C+	2
C- to C	0
D+ to C-	1

Extra Points

Quizzes

Course materials

- Canvas Course Site

  https://canvas.illinois.edu/courses/47527

Lecture videos and ClassTranscribe

- Lecture and discussion will be recorded and accessible at https://mediaspace.illinois.edu/
- ClassTranscribe provides transcripts for these videos

  https://classtranscribe.illinois.edu/home

Ed policy and Gradescope submission

- Students are expected to follow the guidelines on how to use Ed in this course
- Students are expected to follow the guidelines of homework submissions in this course

Our Staff

Instructor: Hongye Liu
Teaching Assistants (sorted by last name):
Ikhyun Cho, Aditya Karan, Simon Kato, Daniel Kiv, Qiaobo Li

Office hours are listed on Canvas main home page will be available in 2nd week.

Our Staff (II)

Course Assistants (sorted by last name): Balajib Balachandran, Shubhib Bhatia, Flora Fan, Philip Nakamura, Yuqing Jian, Jisen Li, Michael Pollack, Nevin Thomas, Frank Xu, Xinyi Ye, Yingjie Yu

Big picture of the content

- Probability and Statistics in action
- What does this course teach?

  Textbook: Forsyth, D. A. "Probability and Statistics for Computer Science," Springer (2018)

- Why are there 4 sections? How are they related?

This field really started with gaming

- We are familiar with flipping a coin or throwing a dice, the result is uncertain!

Head
Or Tail?

Which side
is front?

Life is uncertain so aim for long-term average

- We repeat a lot of experiments and see if there is regularity

Head
Or Tail?

Which side
is front?

Throwing a lot of “coins” for many times in one touch

- Galton board, the Bead Machine

https://www.youtube.com/watch?v=Kq7e6cj2nDw

Probability and Statistics Experiment in action

Simulation of random draw of a picture on computer

- It's the same as throwing a 4-sided die.

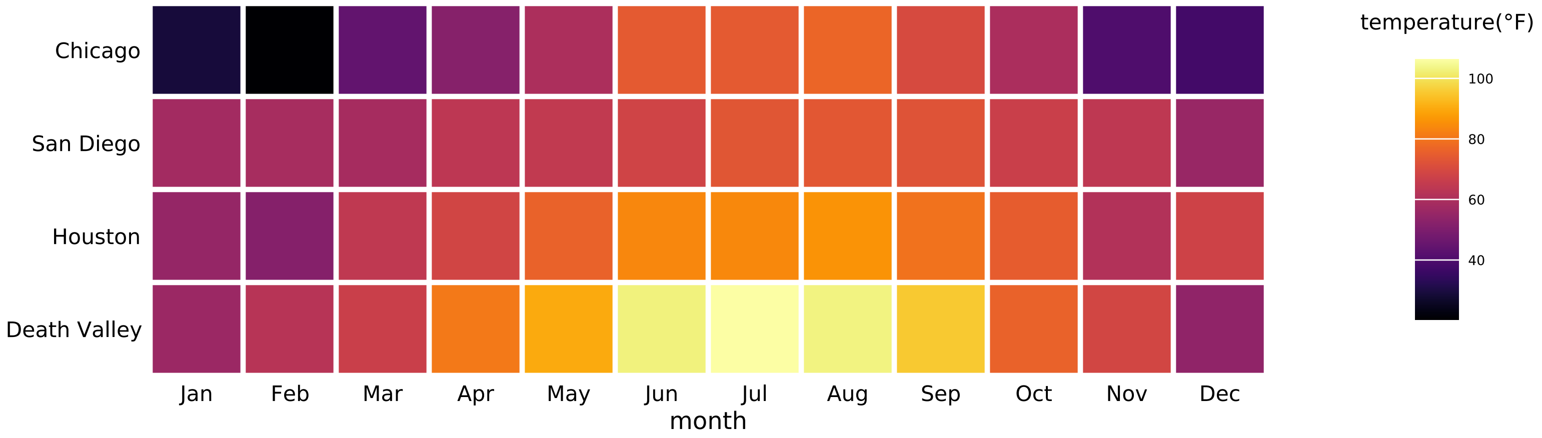
What does this course teach?

- Describing Datasets
  - Summary & visualization
- Probability
- Inference – Statistical Inference
- Tools – Machine Learning tools

Describing datasets (Summary & visualization)

Descriptive & Graphical

Summarization of 4 locations' annual mean temperature by month



Probability

- **Mathematical**

Romeo and Juliet have a date

Each arrives with a delay btw 0 and 1 hour. The first to arrive leaves after 1/4 hour. All pairs of delays are equally likely.

What's the probability that they will meet?

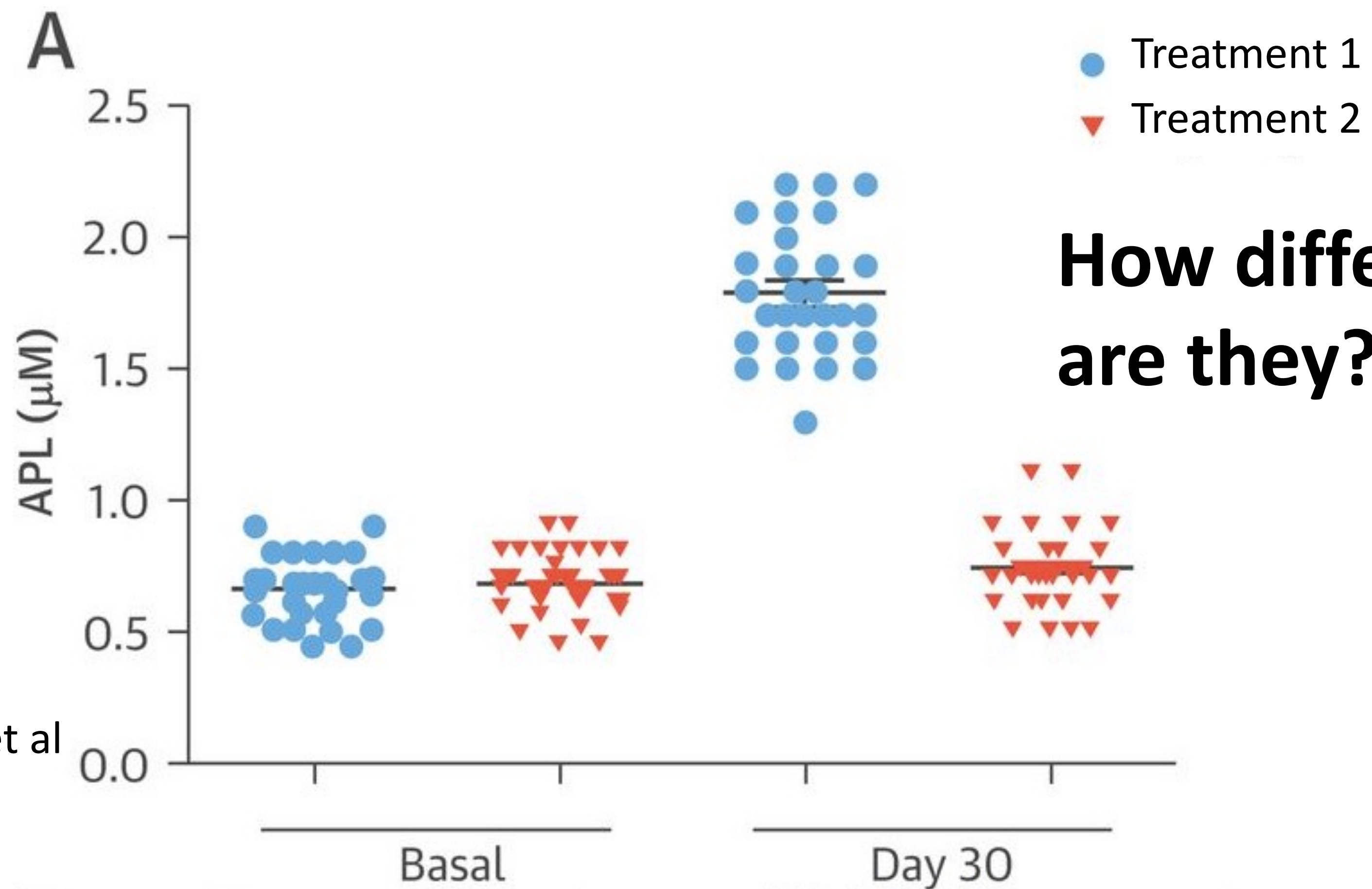
Probability

- **Mathematical**

How many slots are empty on average for a simple hashing table?

Inference

- **Analytical**

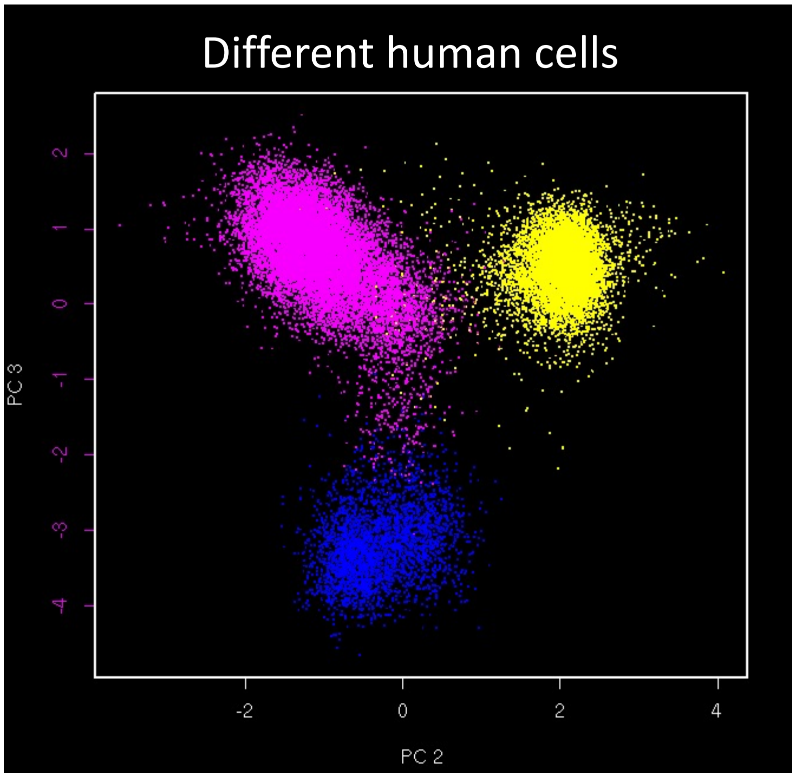


How different are they?

J Fromonot et al
JACC 2016

Tools (Machine learning)

- **Algorithmical**



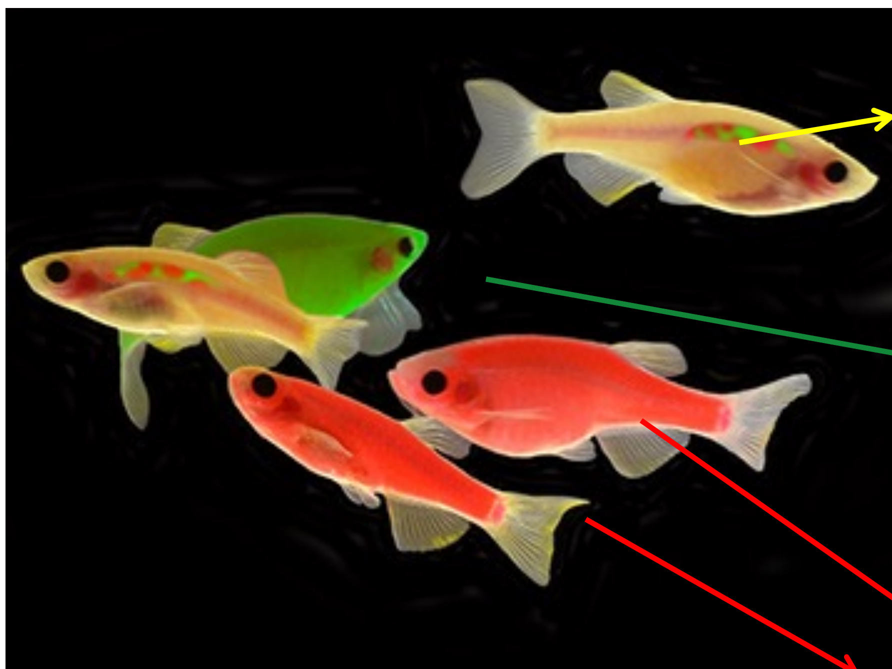
High-dimensional or complex shaped data sets need tools!
Humans are limited in 2-3D.
Machine learning is Highly desired!
Often depends on Statistics.

Why these 4 sections?

- Summary & visualization
  Graphical
- Probability
  Mathematical
- Inference – Statistical Inference
  Analytical
- Tools – Machine Learning tools
  Algorithmical

Why these 4 sections?

- The common thread is **Data.**
- We are doing computer science and so are like these yellow fish



What is special of Data? For Data?

Where were the clown fish in this image located?

what if there is a person standing in front of the fish?

What is special of Data? For Data?

Data is how we see the world.

Context is important for analyzing data!!

Why these 4 sections?

- Real world data is often high dimensional and complex
- These 4 parts of knowledge or techniques are inseparably/organically connected in many real-world applications.

What do we emphasize?

- Mathematical principle
- Critical thinking
- Working with real world data

LECTURE 1

Q. What do you feel about it when we speak of data visualization?

Example 1: Black hole

Constructed image using data collected from many different telescopes' view of the same object

This project received a 3million-dollar award

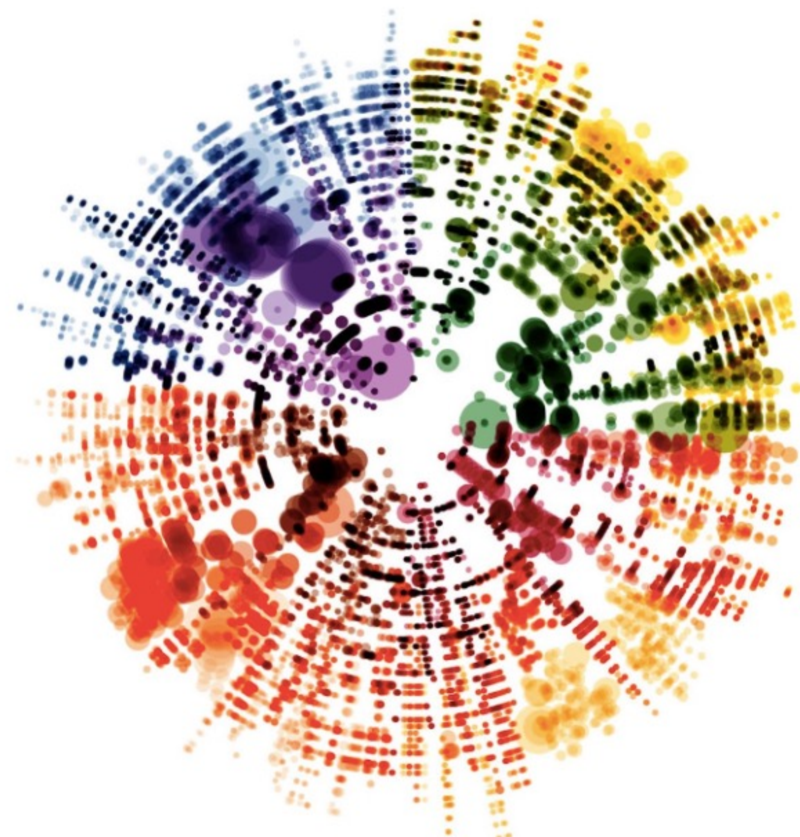
Credit: NASA

Example 2: Four seasons by Vivaldi

Pitch is shown by the distance from center;

Length of the note is the size of dot

Instrument is shown by the color's **shade**



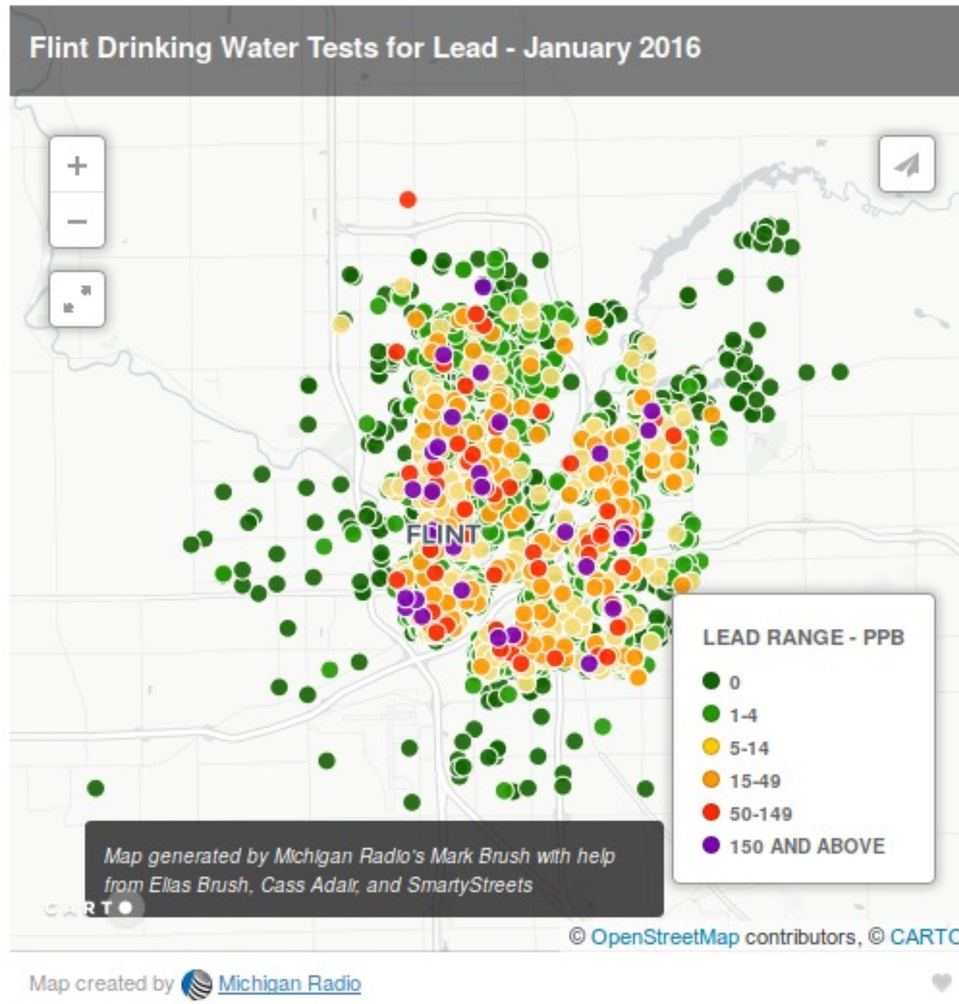
https://medium.com/future-today/off-the-staff-an-experiment-in-visualizing-notes-from-music-scores-58f6ee9f0cef

Example 3: Word cloud

Frequency of words of a document in novel visual presentation

Example 4: GIS map

Color scaled dots show the lead level in water in an area in Michigan



Lecture I: Data Visualization &Summary

- Datasets {x} – a set of N items x_i, i=1…N, each of which is a tuple

Proteins →

Cells ↓

Cell ID	CD45	CD3e	CD19	CD11b	Ki67
1	7.10543765	1.99490875	2.13073358	7.82894178	2.57289058
2	6.5957055	4.65342077	1.62918585	0.88137359	0.88137359
3	6.81991147	1.76259579	4.63429706	2.74452653	0.88137359
4	6.90112651	1.41502227	4.54593607	0.88137359	0.88137359
5	6.75571436	2.87597714	2.18671075	6.72464322	0.91192661
6	7.39538689	2.55285118	4.55845203	1.57273629	0.88137359
7	6.50181654	0.9030504	0.88137359	6.55459538	1.61883699
8	6.60986569	2.1753298	1.52779681	6.44086205	1.5347653
9	6.97651408	2.38246511	1.90249637	3.41580053	1.85303806
10	7.14397512	3.36924119	9.23325502	4.79035059	0.88137359

Each row is a tuple

Lecture I: Data Visualization &Summary

- Convention: columns are the ***features***; the number of features is ***dimension.***

Proteins →

Cells ↓

Cell ID	CD45	CD3e	CD19	CD11b	Ki67
1	7.10543765	1.99490875	2.13073358	7.82894178	2.57289058
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10	7.14397512	3.36924119	9.23325502	4.79035059	0.88137359

Each row is a tuple with dimension =5

Data types

- Categorical
- Ordinal
- Continuous

Data types

- Categorical

  Smoker or non-Smoker, Female or Male etc.

- Ordinal

  Satisfaction (Not satisfied, satisfied, very satisfied)

- Continuous (any real number within a range)

  Temperature

Q. Which of the following data is not categorical?

A. Number of enrolled students in a class
B. Weight of apples in a grocery store
C. Instruments played by an orchestra
D. Type of chemical reagents in a lab
E. A & B

Q. Which of the following data is ordinal

A. Number of legs of the stools sold on Wayfair
B. Capacity of the kettles sold on Amazon
C. Reviews of a book on Amazon (Not the # of stars)
D. Brands of iron boards sold by Target
E. A and C

Simple Visualization of Data

- General principles
- Bar chart
- Histogram
- Conditional histogram

Simple Visualization of Data

- General principles

  Must not mislead or distort;

  Aesthetically pleasing;

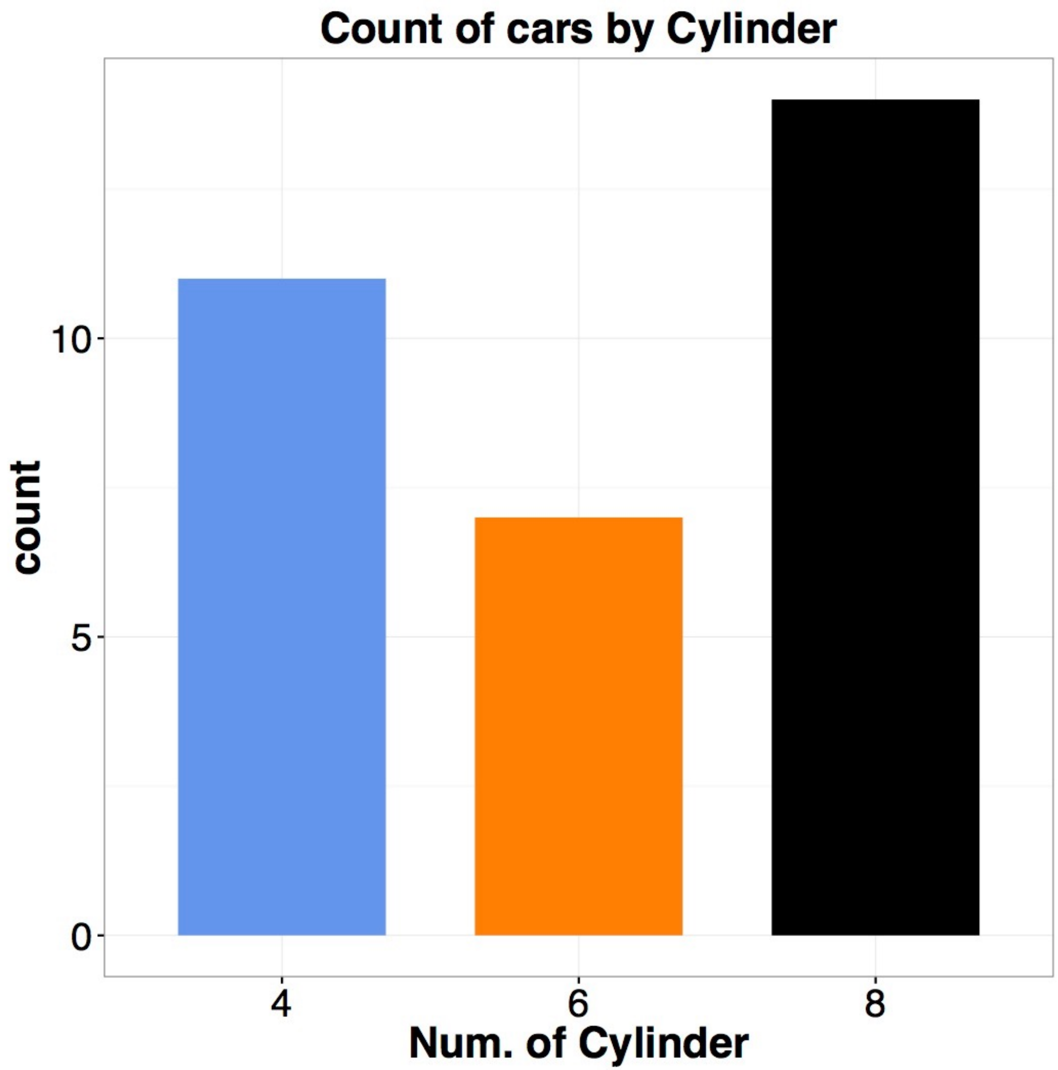
  Clear, Attractive, Convincing;

  Show message/significance.

Simple Visualization of Data

- Bar chart

A set of bars that are organized by categorical or ordinal feature



Data: “mtcars”

An example of good, ugly, bad, wrong

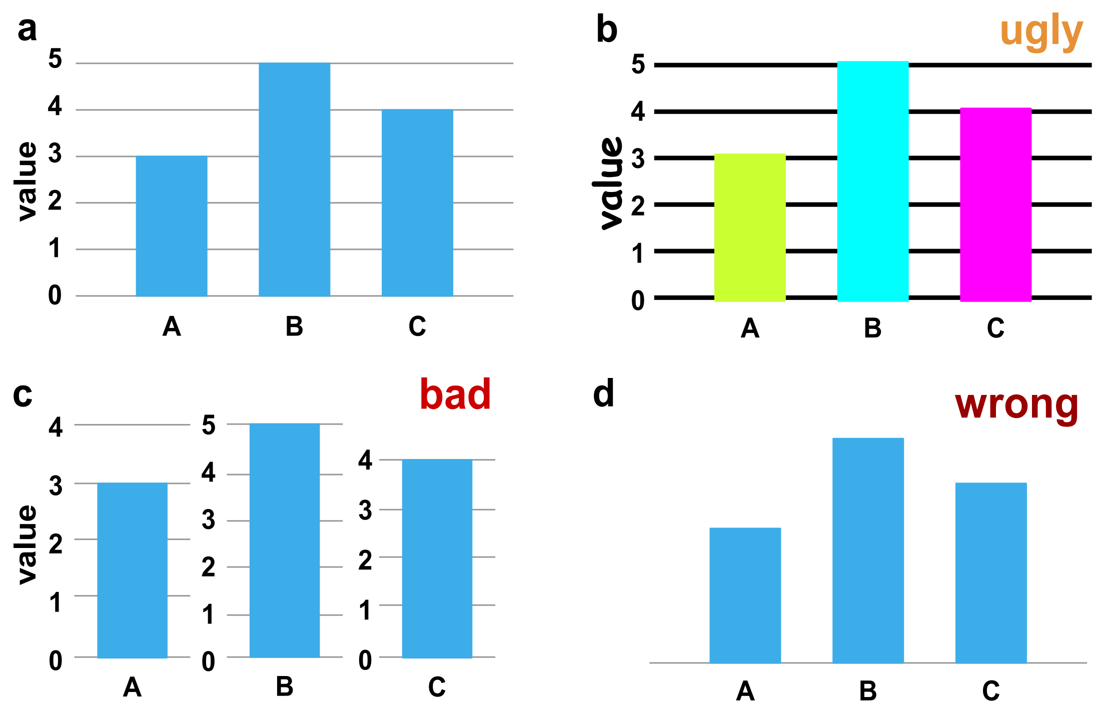
The difference between ***good*, *ugly*, *bad* and *wrong* visualization**

a – good

b – ugly

c – bad

d - wrong



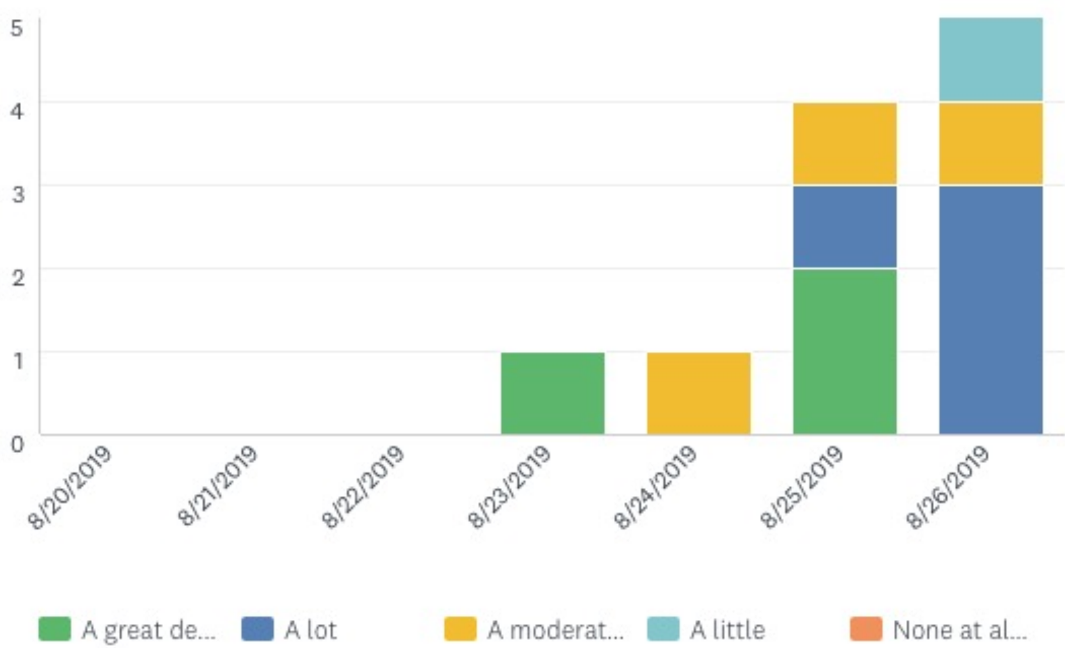
Q: Is this a good bar chart?

Q1 (by day)

Chart Type ▾ Display Options ▾ Trend by... ▾ Zoom ▾

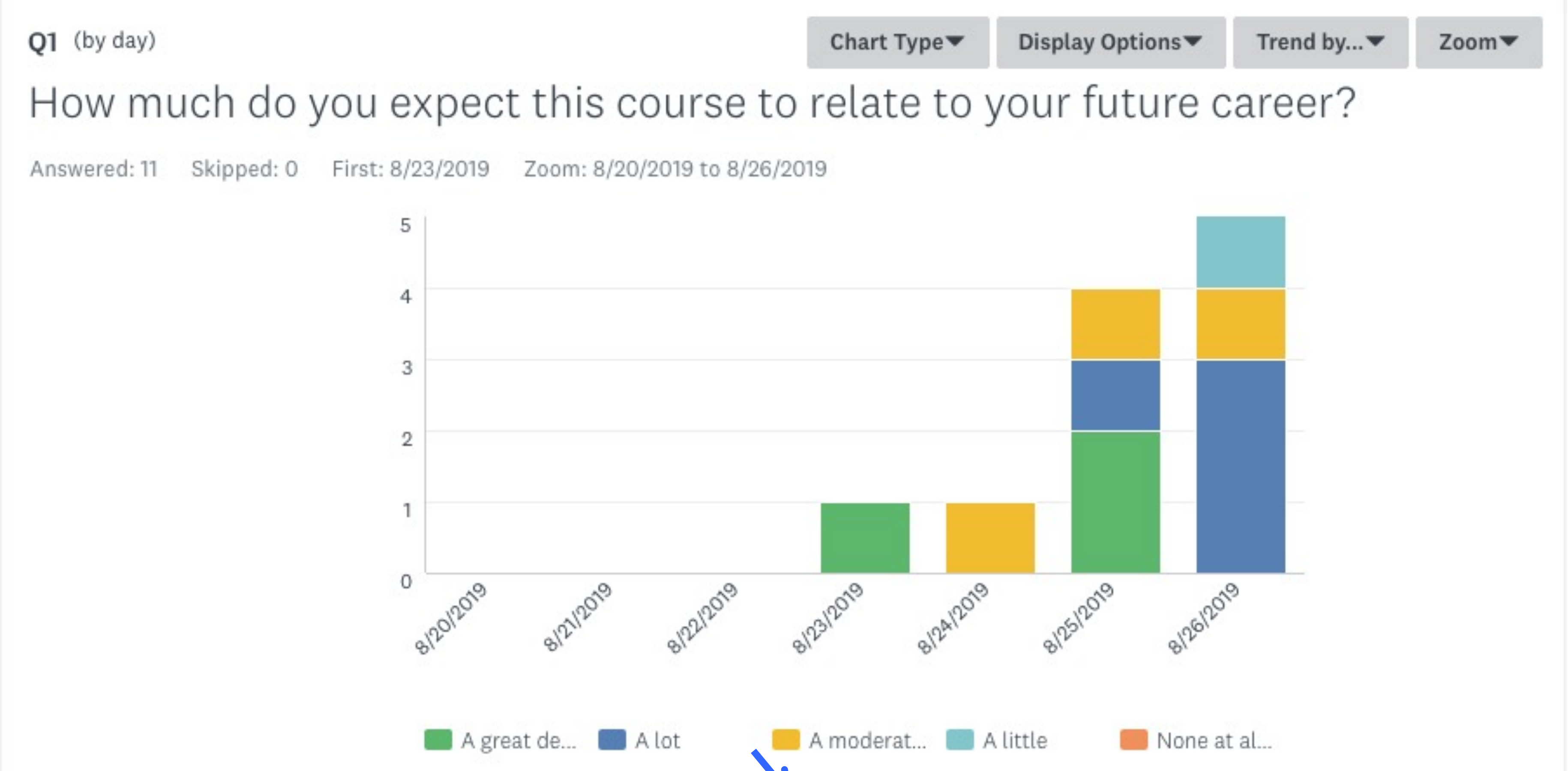
How much do you expect this course to relate to your future career?

Answered: 11 Skipped: 0 First: 8/23/2019 Zoom: 8/20/2019 to 8/26/2019



A. Yes
B. No

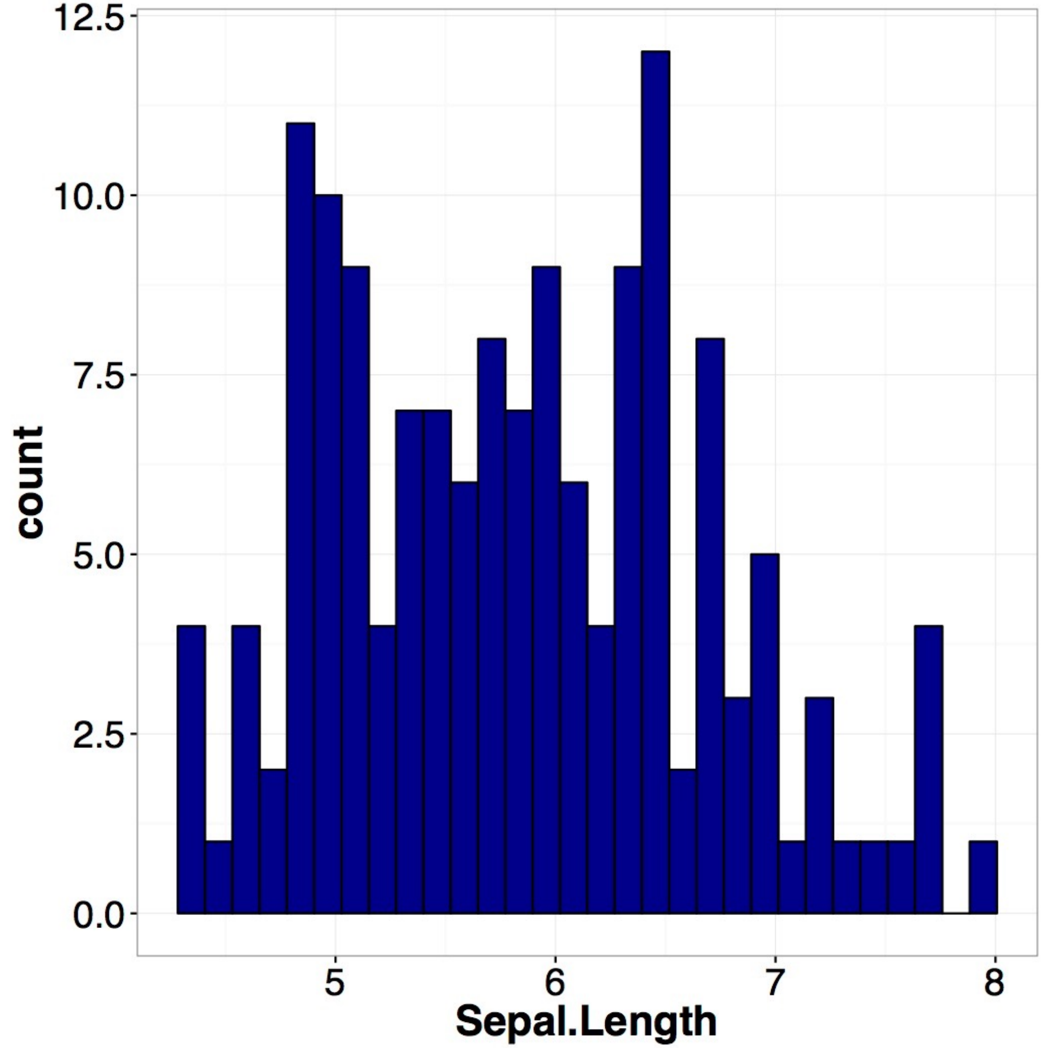
How about using a color scale



Visualizing Data with Histogram

- Histogram

A set of bars that are organized by bins that contains numerical data (discrete or continuous)



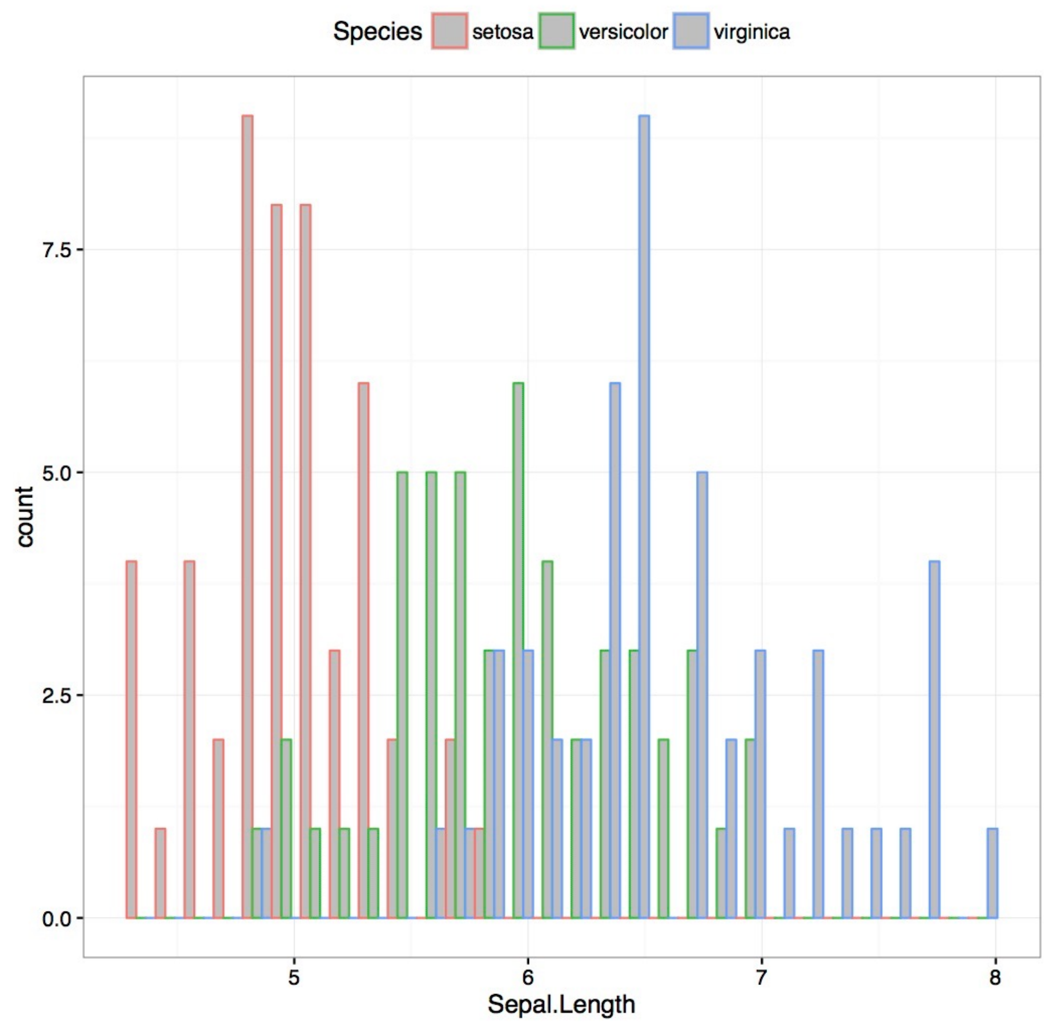
Data: "iris"

Visualizing Data with Histogram (II)

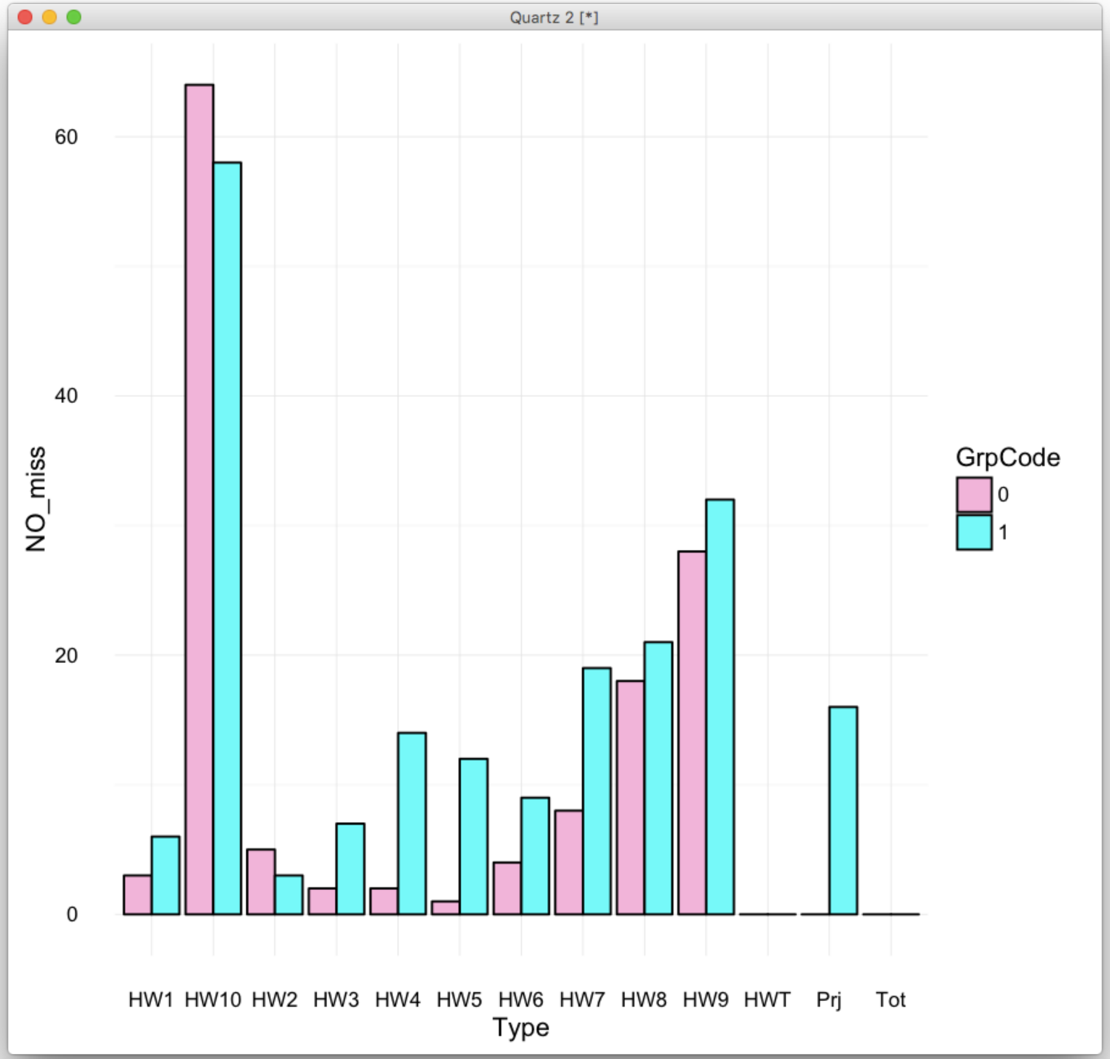
- Conditional histogram

Histogram generated by subsets of the data

Data: “iris”



Which group has the higher total scores?



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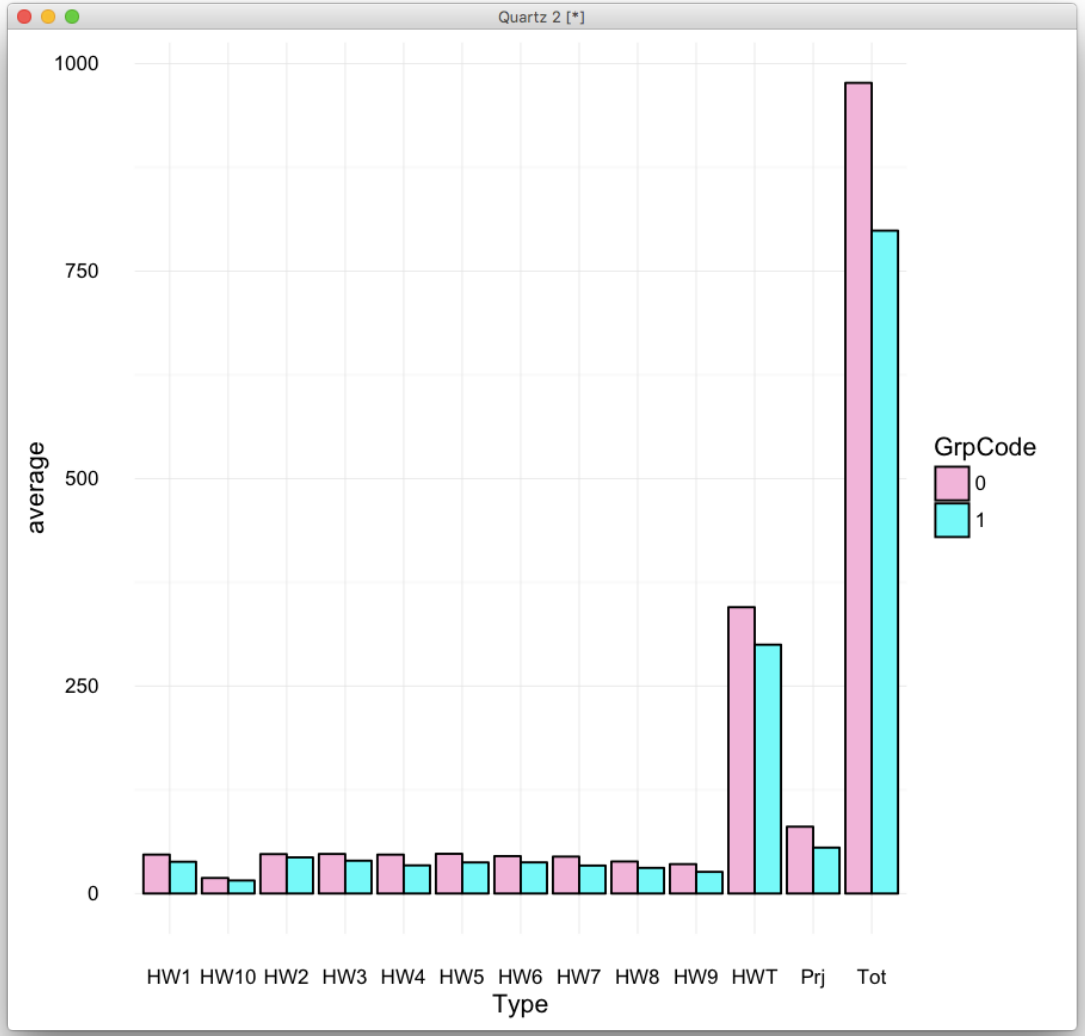

Assignments

- Finish the Orientation module on Canvas
- Submit HW0 to Gradescope to test it
- Work on Quiz0 (optional) to test your Math preparedness.
- Start week1 module on Canvas
- Start graded group discussion #1 about Python

Additional References

- Charles M. Grinstead and J. Laurie Snell "Introduction to Probability"
- Morris H. Degroot and Mark J. Schervish "Probability and Statistics"

See you next time

See you!